

Agricultural Land Abandonment in Nepal

Problem: Identify abandoned agricultural lands (AAL) in Nepal.

Solution (Conceptual): Establish a quantifiable definition of AAL → Develop a method to identify AAL satisfying the definition.

1. Establish a quantifiable definition of AAL

There is no universal definition of AAL. However, “time-based” and “land cover transition” definitions can be found in the literatures.

- Time-based definition: AAL is a land that has been abandoned for a certain period of time.
- Land cover transition definition: AAL is a land that has been converted from agricultural land to non-agricultural land.

Suggested definition for this study

Agricultural land abandonment is defined as arable land (khet or bari) where farming activities have ceased for a minimum of five consecutive years, leading to a transition to grassland, shrubland, or scrub, but which has not been converted to built-up areas, water bodies, or mature forest ([Paudel et al., 2025](#); [Rai et al., 2019](#)).

1.1. Rationale for this definition

- Quantifiable supported by FAO (FAO, 2016)
- Avoids misclassifying temporary fallow land

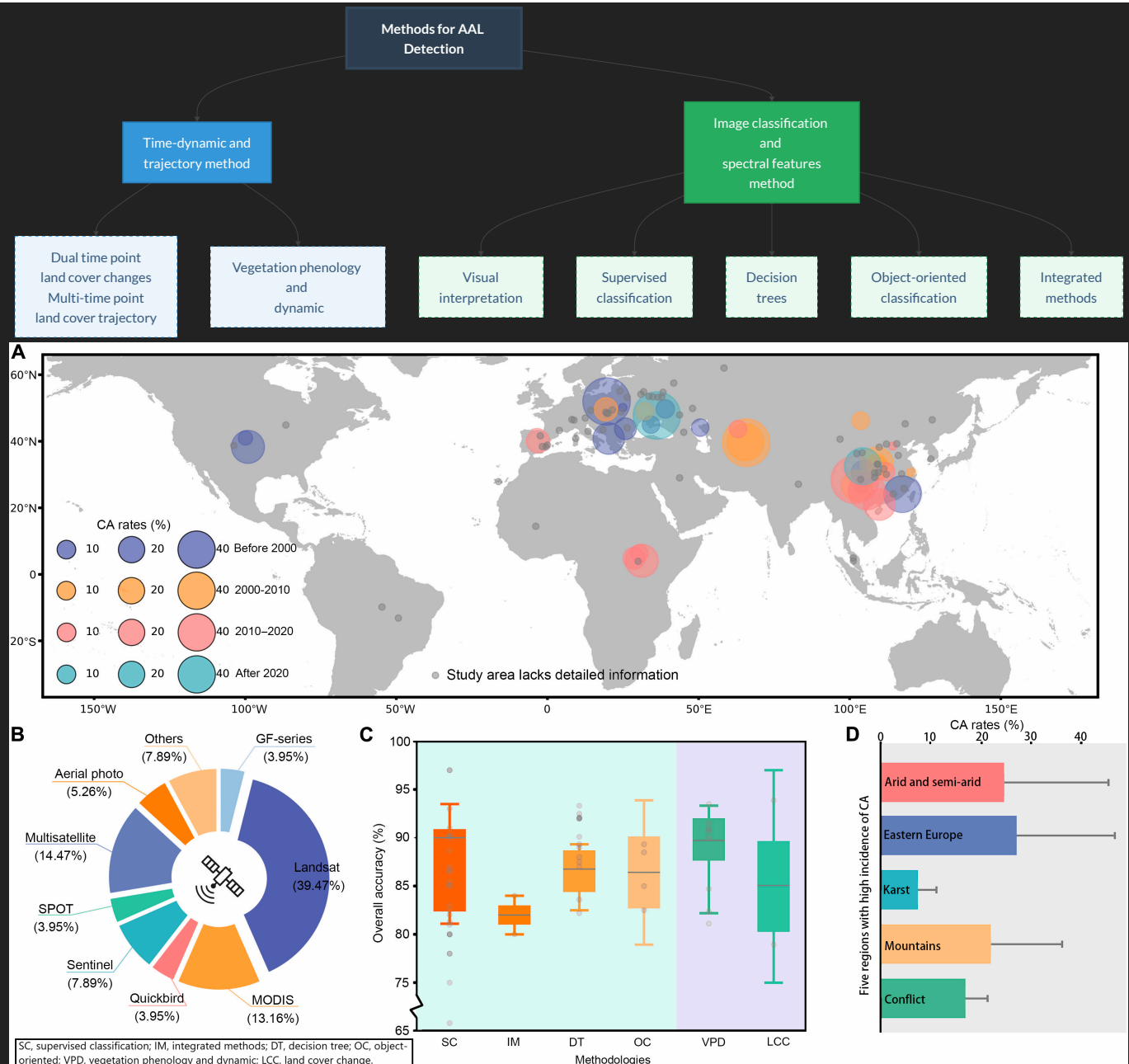
2. Develop a method to identify AAL satisfying the definition

Literatures before 2019 were reviewed from papers cited in ([Goga et al., 2019](#)). Papers published after 2019 were reviewed from google scholar with custom date filter and ([Liu et al., 2025](#)). Review was non-exhaustive till 2026-01-19 11:07.

2.1. Review of existing methods

Based on ([Liu et al., 2025](#)), the methods can be divided into two major categories (Image classification & spectral features method and Time-dynamic & trajectory method) and various sub categories.

- Image Classification and Spectral Features Methods: This includes visual interpretation, supervised classification (SVM, Random Forest), decision trees, object-oriented classification (OBIA), and integrated methods (fusion of techniques).
- Time-Dynamic and Trajectory Methods: This includes dual time-point change detection, multi-time point land cover trajectories (like LandTrendr), and vegetation phenology dynamics.



Liu et al. (2025)

2.1.1 Comparison of Methods

Overall methods	Specific methods	Principles	Strength	Weaknesses	Applicable scales	References
Image classification and spectral features method	Visual interpretation	Abandoned land identification by experts	High accuracy; vector data; user-flexibility [53]	High subjective; high workforces and time costs	Landscape to rural scale	[58,57]
	Supervised classification	Training classification models with abandoned land samples	User-simple; automated	Subjective selection of samples	Landscape to regional scale	[63,143]
	Decision tree	Abandoned land classification based on conditional splitting	Cost-effective; logically sound and interpretable	Vulnerable to spectral interference	Regional to national scales	[22,68]
	Object-oriented classification	Based on physical attributes	Considers both texture and geometric information	Segmentation threshold; requires high computational power	Landscape to rural scale	[9,59]
	Integrated methods	Integration of multiple methods	Combines the strengths of multiple methods to improve both accuracy and fault tolerance	Emphasis on opaque models	Regional to national scales	[67,76]
Time-dynamic and trajectory methods	Dual time point land cover changes	Overlay of cropland data from 2 periods	User-friendly and efficient	Greatly affected by land cover data	Regional and national scales	[43,66]
	Multi-time point land cover trajectory	Moving window approach with land cover pixels	Year-to-year; easily understood by users	Dependent on the land use/cover classification accuracy [84]	Regional to global scales	[106]
	Vegetation phenology and dynamic	Identifying differences in vegetation phenology	High accuracy and sensitivity; adaptable to diverse scenarios	Noise from noncultivated land; requires time-resolved images	Landscape to national scale	[84,94]

Liu et al. (2025)

Table 1: Comprehensive Comparison of Remote Sensing Methods for Abandoned Agricultural Land (AAL) Detection

Study	Approach	Study Area	Data Sources (Spatial & Temporal)	Definition of Abandonment	Methodology / Algorithm	Accuracy / Performance
(Alcantara et al., 2013)	Multi-temporal (MA)	Central & Eastern Europe	MODIS NDVI (250m) Time-series: 2003–2009	Class-based (Static): Land cultivated in the socialist period but covered by successional vegetation (grass/shrubs) during observation.	SVM: Phenological metrics (start of season, amplitude) extracted via TIMESAT.	Moderate: ~49% area-adjusted overall accuracy.
(Estel et al., 2015)	Multi-temporal (MA)	Europe	MODIS NDVI (250m)	Trajectory-based: Land unmanaged/fallow	Random Forest (RF): Annual classification of “Active”	High: ~90.1% average overall accuracy for

Study	Approach	Study Area	Data Sources (Spatial & Temporal)	Definition of Abandonment	Methodology / Algorithm	Accuracy / Performance
			Time-series: 2001–2012	for 2–5 consecutive years.	vs. “Fallow” using normalized NDVI curves.	active/fallow distinction.
(Stefanski et al., 2014)	OBIA / Multi-sensor	Western Ukraine	Landsat (Optical) + ERS SAR (Radar) Years: 1986, 1993, 1999, 2006, 2010	Trajectory-based: Land cover transition (Cropland → Grassland) without recultivation.	OBIA + RF: Superpixel Contour segmentation to delineate fields. Fusion of Optical and Radar.	83.4% Overall Accuracy. 87.2% for 2010 management regimes.
(Löw et al., 2015)	OBIA / Fusion	Kazakhstan	Landsat + RapidEye Time-series: 2009–2014	Trajectory-based: Analyzed within historical field parcels. Abandonment = 5 years of shrub/bare cover.	Decision Fusion: Trained RF and SVM separately on object-based features and fused results via majority voting.	~97% Overall Accuracy. Fusing classifiers significantly improved accuracy over single algorithms.
(Yin et al., 2018)	OBIA / LandTrendr	Caucasus (Russia/Georgia)	Landsat time series (30m) 1985–2015	Dynamic/Trajectory: Land inactive for at least 5 years (based on FAO definition).	RF + LandTrendr: Multi-resolution segmentation into objects. RF for annual probability, then LandTrendr for temporal segmentation.	97% ± 1% Overall Accuracy. Performed substantially better than pixel-level detection (82%).
(Wu et al., 2023)	OBIA / LandTrendr	Fujian Province, China	Landsat + Sentinel-2 (10-30m) 2010–2021	Narrow Sense: Farmland not cultivated for 1 year or more.	RF + LandTrendr: Used existing government farmland plot boundaries. RF for annual probability, then LandTrendr to	High: 87.02% (2018) and 87.50% (2020) validation accuracy.

Study	Approach	Study Area	Data Sources (Spatial & Temporal)	Definition of Abandonment	Methodology / Algorithm	Accuracy / Performance
					identify significant drops.	
(Dara et al., 2018)	Multi-temporal (MA)	Northern Kazakhstan	Landsat time series (30m) 1984–2016	Trajectory-based: 3 consecutive years active followed by 3 consecutive years inactive.	RF + LandTrendr: 3-year moving window for spectral metrics. RF for annual cropland probability, then LandTrendr for trajectory analysis.	89% Overall Accuracy. UA of 93% for the abandonment class.
(Güther et al., 2011)	Texture / Edge Detection	Tenerife (Canary Islands)	RapidEye + Orthophotos 2010	Subtraction Method: (Total Ever-Utilized Area) minus (Current Active Agriculture).	Canny Filter within OBIA: Detects “linear footprints” (terraces/walls) under vegetation to identify long-term fallow.	93.0% Overall Accuracy. Highly effective for abandoned terraces.
(Shixin et al., 2013)	OBIA (Rule-based)	China	SPOT-5 (2.5m/10m) Nov 2010	Land Cover Class: Identification of abandoned land as a specific spectral/structural class.	OBIA: Edge-based segmentation + rules involving spectral signatures, shape, and texture.	78.94% accuracy for Abandoned Class. Outperformed pixel-based methods (~59%).
(Goudin et al., 2015)	OBIA / SVM	Poland / Ukraine Border	Landsat 8 OLI (Single date) Aug 2013	Land Use Distinction: Differentiating “Pasture/Abandoned” from active “Arable” land.	SVM within OBIA: Combined spectral features with object geometry (shape/size).	75% Overall Accuracy for land use. Spatial features were essential for land use separation.
(Suziedely)	Segmentation	Lithuania	Sentinel-2 (Optical)	Verification: Land covered with woody	Mean-Shift Segmentation:	Varied: ~94% (Winter),

Study	Approach	Study Area	Data Sources (Spatial & Temporal)	Definition of Abandonment	Methodology / Algorithm	Accuracy / Performance
te Visoc kiene et al., 2019)			Sept 2017	plants identified by remote sensing and compared with national registry.	Clustering and Kernel density (ColorMapping) to find homogenous scrub regions.	~75% (Summer). Accuracy depends on seasonal vegetation state.

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3. Suggested Methodology

Multitemporal Object-oriented Image Analysis Based on Phenology using landTrendr algorithm

Based on the review of literature ((Goga et al., 2019; Liu et al., 2025) and others), the I suggest the following methodology based on (Dara et al., 2018; Wu et al., 2023; Yin et al., 2018) (mostly (Yin et al., 2018)).

3.1. Rationale for the using the Methodology

- **Precise Temporal Detection:** Unlike phenology-based methods that often provide only a broad window of change (Alcantara et al., 2013), LandTrendr identifies the specific timing of abandonment and potential recultivation (Dara et al., 2018; Wu et al., 2023; Yin et al., 2018).
- **Differentiation of Fallow vs. Abandonment:** Phenology-based annual classifications often confuse temporary fallow years in a crop rotation cycle with permanent abandonment (Dara et al., 2018; Estel et al., 2015). By analyzing the full temporal duration, the proposed method distinguishes these by applying a duration threshold (e.g., labeling abandonment only after 3–5 years of inactivity).
- **Noise Reduction via Trajectory Fitting:** Instead of classifying each year independently, LandTrendr fits a geometrical model to the time series. This smoothes inter-annual noise related to climate or sensor variations, ensuring detected changes reflect real land-cover transitions rather than outliers (Dara et al., 2018; Yin et al., 2018).
- **Superior Accuracy:** Previous studies, such as Yin et al. (2018), indicate that this trajectory-based method—especially when combined with Object-Based Image Analysis (OBIA)—can achieve overall accuracies as high as 97% ± 1%, compared to 82% for standard pixel-based approaches.
- **Training/Validation Sample Requirement:** The sample data requirements for the suggested method (for generating agriculture land probability) focus on stable reference samples rather than year-specific (samples for each year separately) samples. This means it requires significantly less samples compared to agricultural land classification methods like (Estel et al., 2015).

Analogy for Understanding: Phenology-based methods are like a security camera that only takes one photo each year; you can see if the field is green, but you can't tell if the farmer just took a week off. Using LandTrendr + OBIA is like having a 30-

year high-definition time-lapse video of the entire neighborhood; it allows you to ignore “camera flicker” (sensor noise) and see exactly when the farmer walked away and never came back.

3.2. Suggested Methodology in Detail (based on (Dara et al., 2018; Wu et al., 2023; Yin et al., 2018))

This method is designed specifically to address the constraints related to topography of nepal, farming culture, cloud cover, time and other resources limitations, temporal coverage of freely available satellite data and so on.

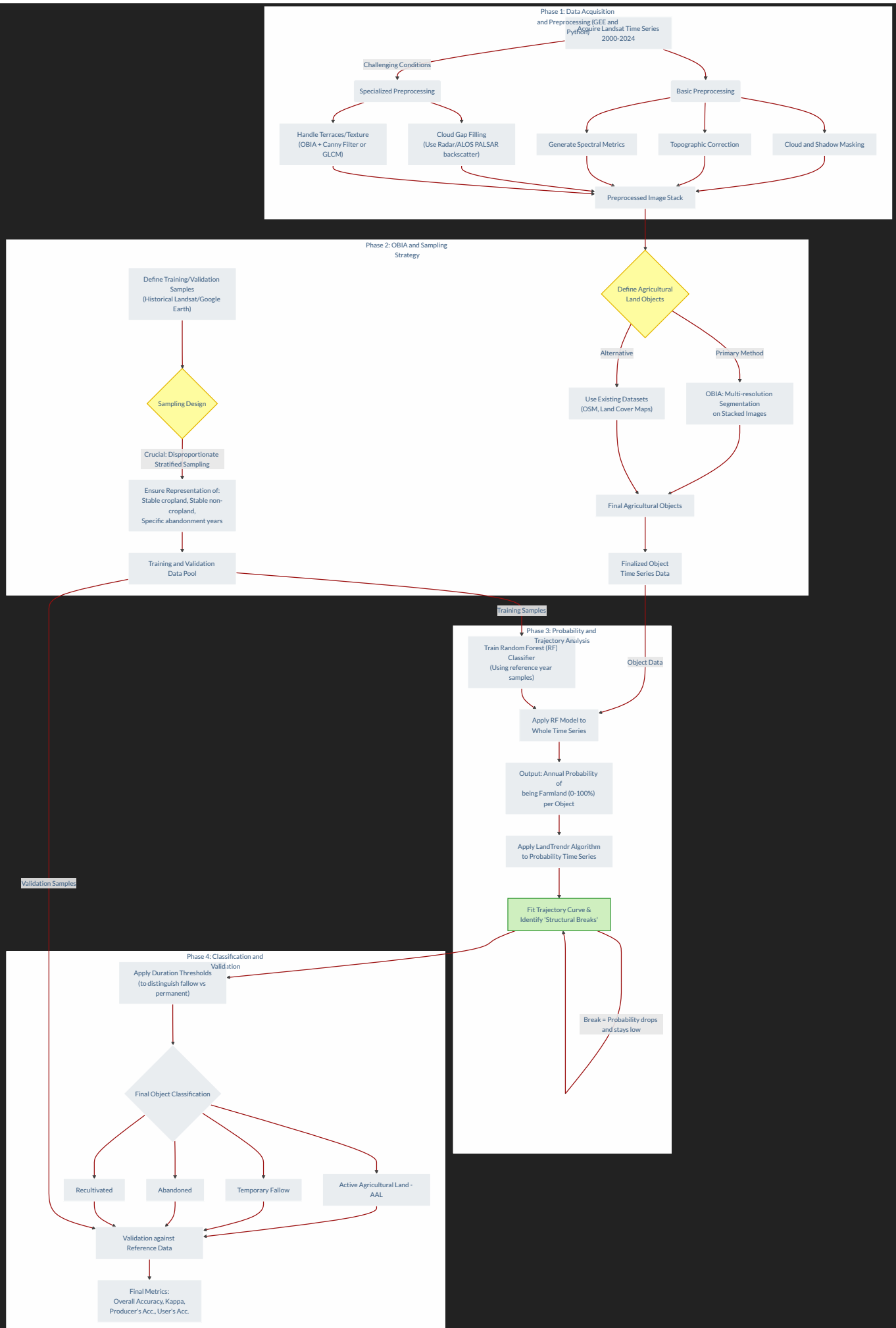
The methodology uses an OBIA landTrendr approach applied to annual class probabilities.

1. Create agricultural land objects: The study will be carried out at object level to reduce “salt and pepper” noise common in pixel-based methods. The multi-resolution segmentation or similar algorithm will be applied to stacked Landsat images to create agricultural land objects (Yin et al., 2018). Instead of using a single image for segmentation, (Yin et al., 2018) applied the algo to stack of landsat images (Aug 1989, Sep 1998, May 2007 and Jul 2015) to create temporally homogeneous objects. They used "bottom-up multi-resolution segmentation algorithm" in eCognition. This approach initiates each pixel as a single segment and merges spatially adjacent segments based on their similarity until the desired scale is met.
 - Alternative 1: Use existing agricultural land from land cover datasets or osm or similar like (Wu et al., 2023).
2. Data Acquisition and Preprocessing: GEE and python will be used for all the analysis if possible to keep the workflow easily repeatable and well documented.
 - Datasets: Landsat (2000-2020/2024)
 - Preprocessing: Cloud Masking, Cloud Shadow Masking, Topographic Correction, Spectral and Textural Metrics
 - Cloud Gap filling: Yusoff et al. (2017) achieved 93% accuracy mapping abandoned paddy and rubber in tropical Malaysia using ALOS PALSAR (L-band Radar). They found that abandoned lands had distinct backscatter signatures compared to active fields.
 - Handelling Terraces: Spectal data alone might fail here. Günthert et al. (2011) used Edge Detection (Canny filter) algorithm combined with OBIA to map abandoned terraces. They found that active terraces have distinct linear textures (walls/furrows) that degrade upon abandonment. Using texture improved their accuracy to 91%. (Wu et al., 2023) used Grey Level Co-occurrence Matrix (GLCM).
 - Training and Validation Samples: Method used by (Wu et al., 2023, 2023; Yin et al., 2018) require training data for an RF classifier to generate annual cropland probability maps.
 - For training, (Dara et al., 2018) used 1,800 sample pixels (900 for stable cropland and 900 for stable non-cropland). For validation, they used 1,290 independent reference samples. This included 50 pixels per stable class and 20–30 pixels per individual year of abandonment/recultivation to ensure the temporal accuracy was captured
 - (Yin et al., 2018) used a larger set of approximately 3,000 Landsat pixel-size training samples, split evenly between active and non-active agriculture
 - (Wu et al., 2023) validated their results using 262 abandoned points in 2018 and 296 points in 2020
 - The sources emphasize disproportionate stratified sampling. This ensures that rare events (like the specific year of abandonment) are adequately represented in the validation pool, as a purely random

sample might mostly hit stable forest or stable farm plots ([Dara et al., 2018](#); [Yin et al., 2018](#)).

- Source of Training Data: Researchers typically use a combination of historical Landsat imagery and high-resolution Google Earth archives to identify “stable” plots (areas that did not change during a specific window) to provide the baseline spectral signature for the classifier

3. Calculate Annual Probability (probability of an object being agricultural land): ([Dara et al., 2018](#); [Yin et al., 2018](#)) used “stable” training sites (areas that did not change). They extracted the spectral signature for those stable sites across multiple years to train the model. Ensure the model is robust enough to handle inter-annual phenological variations and recognize farmland under varying climatic conditions (wet vs dry years).
4. Apply landTrendr to Time Series Probabilities: Feed the annual probabilities values into the landTrendr algorithm. This algorithm fits a trajectory curve to the probability history. It will identify “structural breaks”—specifically, the point where the probability of being farmland drops and stays low.
5. Classify Each Object into “AAL”, “Recultivated”, or “Fallow”: This will be based on threshold like ([Yin et al., 2018](#)) did.
6. Validation: Overall Accuracy, Kappa Coefficient, Producer’s Accuracy, User’s Accuracy



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